Ahmed Othman AI Engineer

ahmedothman1752@gmail.com 01090294387 Egypt-Giza LinkedIn GitHub

Medium Kaggle

Profile

AI graduate with over 2 years of hands-on experience in building **end-to-end machine learning pipelines**, **LLM-based systems**, and **autonomous AI agents** achieving **85%+ accuracy** in real-world settings. Specialized in **Knowledge Graphs**, **RAG systems**, and **Arabic NLP**, with strong skills in **statistical analysis**, **deep learning**, and deploying scalable AI solutions on **cloud infrastructure**.

Education

B.Sc. in Computer Science & Artificial intelligence 2021 – 2025
Nile University Very Good

Graduation Project:

MAYOM - Self-Evolving Knowledge Graph Framework for Enhanced RAG Systems

- Led development of an **autonomous AI agent framework** that enables LLMs to independently build and enrich knowledge graphs.
- Designed **multi-agent orchestration** with specialized roles for knowledge extraction, validation, and graph construction.

Technologies: Llama3, Neo4j, Ollama, Light Graph, Graph RAG, React, LangChain.

Professional Experience

AI Instructor & content creator ,Elevorix Academy Aug 2025 – Present
full time

- Designed and led the *Advanced NLP & Agents* course
- Created hands-on coding projects, tasks, and learning materials
- Trained students in applied AI techniques and real-world problem solving

Co-founder & AI Engineer , Vionex.digital Jun 2025 – Present
part time

- Specialized in agent-based AI systems
- Designed and deployed scalable AI solutions
- Focused on efficiency, automation, and real-world impact

AI Instructor, Inno Code Mar 2025 – present
Part Time

- **Lead** comprehensive AI diploma program covering Python fundamentals through deep learning, NLP, and computer vision applications

Junior Teaching Assistant, Nile University Oct 2024 – Feb 2025
Part Time

- **Teaching** the ''Data Science" reviewing lab tasks with students.

AI Engineer, National Telecommunication Institute Aug 2024 – Oct 2024
Training (80 hours)

- **Collaborated** on Arabic NLP tasks, including Web Scraping, LLM development for language applications.

NLP Engineer, Smsm Mar 2024 – May 2024
Internship

- **Built** web scraping agents with intelligent content extraction and API-driven processing capabilities, while developing Arabic language processing agents for automated content generation and multilingual text analysis workflows.

Data Scientist, Technocolabs Softwares Jul 2023 – Sep 2023
internship

- Led end-to-end data science projects from collection to deployment
- Built predictive models achieving 85%+ accuracy on business forecasting
- Developed automated ML pipelines reducing model training time by 40%

Machine Learning Engineer, Coding Samurai Jun 2023 – Aug 2023
Internship

- A repository of internship projects showcasing my skills in developing machine learning solutions.

- Tools: Python, Flask, Pandas, Scikit-learn, NLTK, Matplotlib/Seaborn, TensorFlow/Keras.

Machine Learning Engineer, TechnoHacks EduTech Official Sep 2023
Internship
- Implemented machine learning models to solve real-world problems.

Entrepreneurship, ITIDA Smart Village Center on Innovation and Creativity Jun 2023
Training
- **Tools:** Persona, Figma, Empathy Map, and BMC.

Internships & Training

AI Engineer, National Telecommunication Institute Aug 2024 – Oct 2024
- **Collaborated** on Arabic NLP tasks, including Web Scraping, LLM development for language applications. Training (80 hours)

NLP Engineer, Smsm Mar 2024 – May 2024
- **Built** web scraping agents with intelligent content extraction and API-driven processing capabilities, while developing Arabic language processing agents for automated content generation and multilingual text analysis workflows. Internship

Data Scientist, Technocolabs Softwares Jul 2023 – Sep 2023
- Led end-to-end data science projects from collection to deployment Internship
- Built predictive models achieving 85%+ accuracy on business forecasting.

Projects

Therapeutic Chatbot with Neo4j Knowledge Graph
- Developed a conversational AI agent for mental health diagnosis using a **107,000-relation knowledge graph**.
- Implemented **multi-turn dialogue management** and **context-aware responses** for realistic patient interaction.
- Achieved better results than **Bio_ClinicalBERT** and **Llama-3_70B** in both single-turn and multi-turn evaluations.
- Built an **agent memory system** to track patient history and deliver personalized responses.

Technologies: Neo4j, Llama-3_70B, TogetherAI API, Gemini API, Agent Memory Systems.

NexusAI: Cognitive Agent Platform with Vector Memory Architecture
- Designed and implemented a **multi-agent system** with specialized agents for chat, document processing, and memory management.
- Built a **central controller** that dynamically routes tasks to agents based on user intent.
- Enabled **collaborative task execution and context sharing** through a custom communication protocol.
- Integrated a **vector-based memory architecture** for seamless, parallel processing and improved performance.

Technologies: Multi-Agent , Agent Communication Protocols, Task Decomposition, Semantic Memory, Vector Embeddings, Intent Recognition, Document Processing Agents, OCR.

Computer Vision Interactive Game & Model Benchmarking Platform Dec 2024
Developed multi-modal AI agent integrating pose detection, facial expression recognition, and object tracking with intelligent agent decision-making for real-time computer vision task routing, implementing model comparison agents (MediaPipe/MoveNet, YOLOv8/SSD) with automated performance benchmarking capabilities.
•**Technologies:** OpenCV, TensorFlow, MediaPipe, YOLOv8, Agent Orchestration

Customer Churn Prediction & Retention System

- Built a machine learning pipeline using **XGBoost**, achieving **~0.93 ROC AUC** in churn prediction.
- Simulated **risk segmentation and retention**, estimating **$174,570 in net savings** by targeting 253 high-risk customers.
- Deployed a **scalable prediction function** for integration with CRM systems.

Technologies: Python, Pandas, Scikit-learn, XGBoost, SHAP, Imbalanced-learn, Joblib

Recommendation System

Built a robust recommendation engine using the **MovieLens 20M dataset** with two approaches:

- **Memory-Based Collaborative Filtering (Pearson & Cosine)** for explainable, item-to-item recommendations.
- **Model-Based Collaborative Filtering (SVD)** achieving **RMSE ~0.73** & **MAE ~0.55** on 250k+ users & 40k+ movies.

- **Technologies:** Python, Pandas, Scikit-learn, Scikit-Surprise, Matplotlib

Soccer Jersey Generation using Adversarial Neural Networks Aug 2024

Focused on generating unique and innovative soccer jersey designs
•**Models:** Generative Adversarial Networks(GANs)
•**Tools:** TensorFlow/Keras - StyleGAN - Beautiful Soup - Selenium - Image Augmentation - Flask - JS

Music Analytics & Recommendation System

- Analyzed Billboard HOT 100 data to predict hit songs using audio features
- Built recommendation engine using collaborative filtering and content-based methods
- Achieved 78% accuracy in hit prediction using Neural Networks and feature engineering

- **Tools:** LibROSA -NN - Feature Extraction - Billboard API - Scikit-learn - Flask - JS

Predictive Analytics Projects

- Big Mart Sales Prediction: End-to-end ML pipeline achieving 89% accuracy
- Gold Investment Dashboard: Real-time financial data visualization with Streamlit
- **Technologies:** Python, Scikit-learn, Neural Networks, Streamlit, APIs

OCR & Document Processing Application

- Built a document processing agent with **intelligent preprocessing and text extraction** features.
- Implemented a **multi-stage agent workflow** for document analysis and quality assessment.
- Deployed the solution on **Streamlit Cloud** using a scalable, cloud-based agent architecture.

Technologies: Python, OpenCV, Tesseract, Streamlit, Docker, Agent Pipelines

Visual Venture Repo

- A curated collection of my advanced computer vision projects, showcasing my deep passion for the field. These projects leverage state-of-the-art deep learning architectures and cutting-edge techniques to tackle complex real-world CV challenges.
- **Tools & Frameworks:** TensorFlow, PyTorch, OpenCV, MMDetection, Detectron2, Albumentations, ONNX, NVIDIA TensorRT, Scikit-learn, and more.
- **Models & Architectures:** CNN, ViT, Swin Transformer , EfficientNet, ResNeXt, MobileNet, DenseNet, YOLO, Faster R-CNN, Mask R-CNN, U-Net, SegFormer, and more.

NLP Projects Repo

- A dedicated collection of my advanced NLP projects, highlighting both my academic focus and practical expertise in deep learning, language modeling, and text analytics. These projects tackle a range of NLP challenges.

- **Tools & Frameworks:** TensorFlow, PyTorch, Hugging Face Transformers, AllenNLP, SpaCy, NLTK, FastText, Fairseq, Seq2Seq,and more.
- **Models & Architectures:** BERT, GPT-3, LLaMA, T5, BART, XLNet, RoBERTa, ALBERT, Electra, LSTM, BiLSTM-CRF, Transformer-based Sequence Models, and more.

Freelance Projects

AI Meeting Optimizer & Action Item Tracker Using n8n Jul 2025

- Developed an **automated meeting transcription platform** using **n8n** and **Azure OpenAI API**, handling **25MB+ audio files**.
- Integrated with **Google Sheets, Slack, and Gmail APIs** for smart task assignment and progress tracking.

Key Technologies: n8n, Azure OpenAI, REST APIs, Slack, Postman.

Art Gallery Project Feb 2025 – Present
Web Developer

- Developed a React-based digital gallery platform with integrated AI chatbot using Supabase and Firebase for secure database, delivering optimized performance solution.

Skills

- **Programming:** Python, Java, C++.
- **AI/ML:** TensorFlow, PyTorch, Scikit-learn, LangChain, AutoGen, CrewAI
- **Data Science:** Pandas, SQL, MongoDB, Neo4j
- **MLOps:** AWS, Docker, MLflow, FastAPI

Certificates

Machine Learning Specialization | Training Certificate at NTI | Deep Learning Specialization | Natural Language Processing Specialization | Technocolabs Certificates & Letter of Recommendation | Large Language Model Operations (LLMOps) Specialization